

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: INTERNET PROGRAMMING

COURSE CODE: CSA4305

COURSE OUTCOME COVERED

CO 5: Construct interoperable web services by leveraging JAX-RPC, WSDL, SOAP, and XML Schema for seamless data exchange across distributed systems. (L4, L5)

Assessment Tool : Industry Problem Task

Weightage: 40%

QUESTIONS:

Total Marks: 30

1.A hospital management system requires a **SOAP-based web service** to manage patient information such as registration, appointment scheduling, and medical records. The system must ensure secure, structured, and interoperable communication between different hospital departments.

Design a **SOAP-based web service** for the hospital system. Explain the structure of SOAP messages (Envelope, Header, Body), define the operations involved (e.g., add patient, get records), and illustrate how data is exchanged between client and server. Justify the use of SOAP in this scenario considering reliability and security.

2.An enterprise application needs to expose its web services so that external systems can understand how to interact with them. The organization plans to use **WSDL (Web Services Description Language)** to describe the service.

Design the **WSDL structure** for a web service that provides operations such as retrieving and updating data. Explain the key components of WSDL (types, message, portType, binding, service), and illustrate how they define the service interface and communication details. Provide a clear structure and justify its importance in service integration.

3. A company is experiencing issues in its SOAP-based communication system where requests are not processed correctly, and responses are either delayed or incorrect.

Analyze the SOAP communication process and identify possible issues such as message formatting errors, incorrect namespaces, or endpoint problems. Propose a debugging

approach to resolve these issues, including validation of SOAP messages and error handling techniques. Explain how the corrected system ensures reliable communication.

Code of Conduct:

I P Madhavi/192372133 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:Madhavi

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding ; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

1. SOAP-Based Web Service for Hospital Management System

Introduction

SOAP (Simple Object Access Protocol) is an XML-based protocol used for communication between applications over HTTP/HTTPS. It provides secure, reliable, and structured communication between different hospital departments such as Registration, Appointment, Billing, Pharmacy, and Medical Records.

Hospital SOAP Web Service Flow Chart

Patient / Client

|



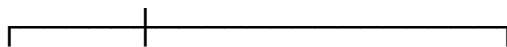
SOAP Request (XML Message)

|



SOAP Web Service

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Registration Appointment Medical Records Billing

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Database Processing

|



SOAP Response (XML)

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Patient / Client

SOAP Message Structure

SOAP messages contain three main parts.

SOAP Element	Explanation
Envelope	Root element that contains the complete SOAP message.
Header	Contains authentication details, security tokens, timestamps, and transaction information.
Body	Contains the actual request or response data exchanged between client and server.

SOAP Message Flow Chart

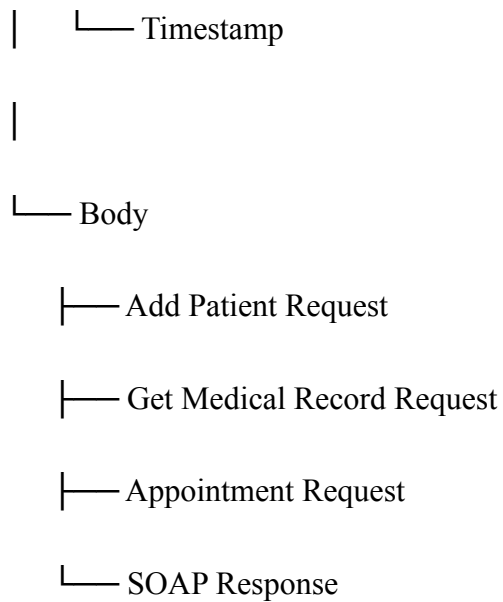
SOAP Envelope

|

| — Header

| | — Authentication Token

| | — Security Information



SOAP Operations

Operation	Purpose
addPatient()	Registers a new patient in the hospital system.
scheduleAppointment()	Books an appointment with a doctor.
getMedicalRecord()	Retrieves patient medical history and reports.
updatePatient()	Updates patient information.
deletePatient()	Removes patient information if required.

Client-Server Communication

Client



| SOAP Request



SOAP Server

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| Process Request



Hospital Database

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| Retrieve/Store Data



SOAP Server

|

| SOAP Response



Client

Why SOAP is Used in Hospital Management?

Feature	Justification
Security	Supports WS-Security for authentication and encryption of patient data.
Reliability	Ensures reliable delivery of messages between departments.
Interoperability	Allows communication between systems developed in different programming languages.
Standard XML Format	Provides structured and platform-independent communication.

Advantages of SOAP

- Secure communication using XML encryption and WS-Security.
- Reliable message delivery.
- Platform and language independent.
- Suitable for enterprise and healthcare applications.

Conclusion: SOAP is ideal for hospital management because it securely exchanges sensitive patient information and ensures reliable communication between different departments.

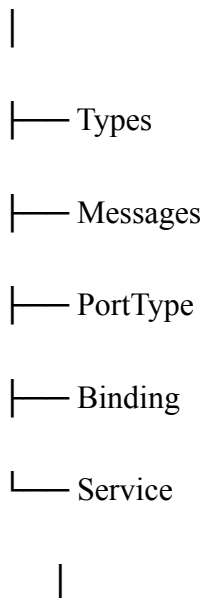
2. WSDL (Web Services Description Language) Structure

Introduction

WSDL (Web Services Description Language) is an XML document that describes how a web service works. It defines the available operations, input/output messages, communication protocol, and service location.

WSDL Architecture Flow Chart

WSDL Document





Client Accesses Web Service

WSDL Components

Component Purpose

Types	Defines XML data types used in requests and responses.
Message	Defines input and output messages exchanged.
PortType	Lists operations offered by the web service.
Binding	Specifies communication protocol (SOAP/HTTP).
Service	Provides the endpoint URL where the service is available.

WSDL Structure Flow Chart

WSDL

|

|— Types

| |

| └— Patient Data, Record Data

|

|— Messages

| |— Request Message

| └— Response Message

|

└─ PortType

| └─ getPatientRecord()

| └─ updatePatientRecord()

|

└─ Binding

| └─ SOAP over HTTP

|

└─ Service

 └─ Hospital Service URL

Example Operations

Operation Input / Output

getPatientRecord() Input: Patient ID → Output: Patient Details.

updatePatientRecord() Input: Updated Patient Data → Output: Success Status.

Communication Process Using WSDL

Client

|

▼

Reads WSDL File

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Understands Operations



Creates SOAP Request



Web Service



SOAP Response

Importance of WSDL

Importance	Explanation
Service Description	Explains available operations to clients.
Interoperability	Different applications can communicate using a common description.
Automatic Client Generation	Development tools generate client code from WSDL.
Standard Communication	Defines protocol, message format, and endpoint clearly.

Conclusion: WSDL acts as the contract between the client and web service, making integration easy and standardized.

3. SOAP Communication Process and Debugging

Introduction

SOAP communication follows a request-response model. If SOAP requests or responses are incorrect, communication fails. Errors may occur because of invalid XML, namespace issues, endpoint configuration, or server faults.

SOAP Communication Flow Chart

Client

|

| SOAP Request

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SOAP Server

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| Validate XML

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▼

Business Logic

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▼

Database

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▼

SOAP Response

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Client

Common SOAP Communication Issues

Issue

Reason

Message Formatting Error Invalid XML syntax or missing SOAP elements.

Incorrect Namespace Wrong namespace URI prevents message processing.

Endpoint Error Wrong service URL or unavailable server.

Authentication Failure Invalid security token or credentials.

SOAP Fault Server returns an error message describing the problem.

SOAP Error Identification Flow Chart

SOAP Request

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Check XML Format

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Check Namespace

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Check Endpoint URL



Check Authentication Header



Process Request Successfully

Debugging Approach

Step	Description
1. Validate XML	Ensure SOAP request follows correct XML syntax.
2. Verify Namespace	Check namespace URI matches the WSDL definition.
3. Verify Endpoint	Confirm correct service URL and server availability.
4. Check Headers	Validate authentication token and security information.
5. Analyze SOAP Fault	Read fault code and fault string returned by the server.

SOAP Fault Structure

SOAP Fault



└─ Fault Code

└─ Fault String

└─ Fault Actor

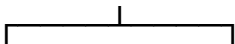
└─ Detail

Error Handling Flow Chart

Receive SOAP Request



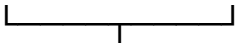
Validate Request



Valid Invalid



Process SOAP Fault Message



Send SOAP Response

How Corrected SOAP System Ensures Reliability

Technique	Benefit
XML Validation	Prevents malformed SOAP messages.
Correct Namespace	Ensures proper message interpretation.
Correct Endpoint	Routes requests to the intended web service.
SOAP Fault Handling	Provides meaningful error information.
Authentication & Security	Protects communication and sensitive data.

Conclusion

By validating SOAP messages, verifying namespaces and endpoints, and handling SOAP faults properly, the communication system becomes secure, reliable, and interoperable, ensuring accurate request processing and correct responses between client and server.